Johnny Dunn

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SUMMARY

Full-stack engineer in end-to-end applied AI/ML (natural language processing / computer vision): RAG pipelines, LLM fine-tuning, MCPs, deep learning, GPU-accelerated training, quantization, embedded development / edge inference, evaluation harnesses, IVRs.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer
Mar 2023 – Present

Independent Consulting (Contract) — Los Angeles, CA

- Acting tech lead building edge AI vision on embedded SBC in Go for Aye Eye Global. Multi-cam streams, WebRTC, websockets, deep learning deployment, quantized LLMs for voice interface, dashboard in React, TypeScript, Vite.
- Improved AI text detection accuracy from 55% to 71% for Grapple Research, finetuned GPT-3.5, deployed with Docker, Supabase on AWS. Built scrapers for training, reduced costs by 64% with document summarization / caching.
- Developed semantic similarity matching tool for ImmenseX (stealth SaaS), on Weaviate / Neo4j RAG (hybrid vector / graph search), on GCP / Linux, with vLLM serving open weights models. Reduced prompt writing time by 60%.
- Built a Python ML pipeline on Databricks / Spark for Linkby, orchestrated with Airflow; automated feature engineering and model training from Snowflake ad data, running RF, SVM, Naive Bayes, and LR in parallel with A/B testing.

Senior Blockchain Engineer
Nov 2021 – Feb 2023

Tilting Point — New York, NY

- Acted as blockchain tech lead; created ERC-20/721 upgradable / proxy smart contracts, built REST / gRPC APIs for marketplace with SSO and containerization, and airdropping tool. Rescued $1M+ in MATIC reverse engineering Gnosis safe.
- Architected Go microservices streaming on/off-chain events via Kafka for Unity SDKs in 4+ games; 1000+ TPS analytics.

Software Engineer 2
Jun 2020 – Apr 2021

eBay Inc. — New York, NY

- Built REST APIs in Scala / Spring-Boot streaming new ads in real time via Kafka from selected item-option clicks (under 1s latency).
- Handled E2E MLOps infrastructure (CI/CD with K8s, Jenkins, MLflow); enabled teams to deploy models to sandboxes exposing data-science-driven statistics. Accelerated deep learning model iterations by 50%; built scalable scrapers verifying data integrity.

Machine Learning Engineer
Jan 2018 – May 2020

Independent Consulting (Contract) — New York, NY

- Achieved 77% sentiment analysis accuracy (random forest) trained on 10K texts for HereAfterLegacy.ai; built MVP with scrapers for TV scripts, recreated digital persona of characters, deployed on Google Cloud Platform.
- Trained neural style transfer models with REST API exhibit applying film-inspired filters to selfies for Edelman.
- Created user research, animated wireframes in Figma, and functional demo for MVP at Notebrush.

Cognitive Implementation Engineer
Jun 2017 – Dec 2017

Amelia (formerly IPsoft Inc.) — New York, NY

- Built AI assistant enterprise modules (PII redaction for HIPAA, SOC 2 compliance, Agile practices) for F500 medical suppliers.

OPEN-SOURCE PROJECTS

Paracosm – AI agent swarm simulation engine as structured world model, built on AgentOS. github.com/framerslab/paracosm
AgentOS – TypeScript agent runtime with multimodal RAG / voice; 680+ stars; 85.6% on LongMemEval-S. github.com/framerslab/agentos
tenets – MCP server for coding agents to aggregate context for prompts with neural ranking, fully offline. github.com/jddunn/tenets
Restless – Hierarchical attention network model for anti-malware; 93% accuracy, ClaMP dataset. github.com/jddunn/restless

EDUCATION

Bachelor of Fine Arts in Game Design — Parsons School of Design, 2017

AWARDS

5th Place, AT&T National VR and AR Challenge, with THiNKIMMERSIVE team (npe100.com / mobile app), 2017.

SKILLS

Languages: Python, TypeScript, JavaScript, Go, Scala, Solidity, SQL, Bash
Agentic & LLM: LangGraph, MCP, DSPy, LlamaIndex, Mastra, RAG, PEFT / LoRA, vLLM, RAGAS, DeepEval, OpenAI, Anthropic
ML & Deep Learning: PyTorch, TensorFlow, Keras, Transformers (BERT, Llama, Mistral, Whisper), scikit-learn, XGBoost, CUDA
Backend: FastAPI, Django, Node.js, Express, NestJS, Spring Boot, gRPC, GraphQL, REST, RabbitMQ, Stripe
Frontend: React, Next.js, Vue, Nuxt, Redux, Tailwind, React Native, Vite, D3.js, three.js
Databases & Vector Stores: PostgreSQL, Supabase, MongoDB, DynamoDB, Redis, Pinecone, Weaviate, Neo4j, Elasticsearch, SQLAlchemy
Cloud & DevOps: AWS (Bedrock, SageMaker, Lambda, EC2), GCP, Terraform, Docker, Kubernetes, Spark, Airflow, Kafka, Snowflake, Databricks, MLflow
Tools: Unity, Jenkins, GitHub Actions, PostHog, Electron, Capacitor, Android Studio, Selenium, Puppeteer